hdoc day 35 by trishil chauhan

Aim

Write a menu-driven C program using a variable-length square integer array and separate functions to test distinct main diagonal elements, calculate their sum, and display alternating diagonal traversal.

C Program

```
#include <stdio.h>

void readMatrix(int n, int a[n][n]) {
    printf("Enter %d matrix elements:\n", n * n);
    for (int i = 0; i < n; i++)
        for (int j = 0; j < n; j++)
            scanf("%d", &a[i][j]);
}

int distinctDiagonal(int n, int a[n][n]) {
    for (int i = 0; i < n; i++)
        for (int j = i + 1; j < n; j++)
            if (a[i][i] == a[j][j])
                return 0;
    return 1;
}

long long diagonalSum(int n, int a[n][n]) {
    long long sum = 0;
    for (int i = 0; i < n; i++)
        sum += a[i][i];
    return sum;
}

void diagonalTraversal(int n, int a[n][n]) {
    printf("Diagonal traversal: ");
    for (int s = 0; s <= 2 * n - 2; s++) {
        int low = (s >= n) ? s - n + 1 : 0;
        int high = (s < n) ? s : n - 1;
        if (s % 2 == 0) { /* upward-right */
            for (int i = high; i >= low; i--)
                printf("%d ", a[i][s - i]);
        } else {          /* downward-left */
            for (int i = low; i <= high; i++)
                printf("%d ", a[i][s - i]);
        }
    }
    printf("\n");
}

int main(void) {
    int n, choice;
    printf("Enter order of square matrix: ");
    if (scanf("%d", &n) != 1 || n <= 0) {
        printf("Invalid matrix order.\n");
        return 1;
    }
    int a[n][n];
    readMatrix(n, a);

    do {
        printf("\n1. Check distinct main diagonal elements\n");
        printf("2. Sum of main diagonal elements\n");
        printf("3. Diagonal traversal\n4. Exit\n");
        printf("Enter choice: ");
```

```
        if (scanf("%d", &choice) != 1) {
            printf("Invalid input.\n");
            return 1;
        }
        switch (choice) {
            case 1:
                printf(distinctDiagonal(n, a)
                    ? "All main diagonal elements are distinct.\n"
                    : "Main diagonal elements are not distinct.\n");
                break;
            case 2:
                printf("Main diagonal sum = %lld\n",
                       diagonalSum(n, a));
                break;
            case 3:
                diagonalTraversal(n, a);
                break;
            case 4:
                printf("Exiting program.\n");
                break;
            default:
                printf("Invalid choice. Try again.\n");
        }
    } while (choice != 4);
    return 0;
}
```

Sample Run

```
Enter order of square matrix: 3
Enter 9 matrix elements:
1 2 3
4 5 6
7 8 9

Choice 1: All main diagonal elements are distinct.
Choice 2: Main diagonal sum = 15
Choice 3: Diagonal traversal: 1 2 4 7 5 3 6 8 9
Choice 4: Exiting program.
```

Result

The program checks uniqueness of the main diagonal, computes its sum, and traverses successive anti-diagonals alternately upward-right and downward-left.